

- **Rapid application deployment:** Containers include the minimal runtime requirements of the application, which reduces their size and allows them to be deployed quickly.
- **Portability across machines:** An application and all its dependencies can be bundled into a single container that is independent from the host version of the Linux kernel, platform distribution or deployment model. This container can be transferred to another machine that runs Docker, and executed there without compatibility issues.
- **Version control and component reuse:** Users can track successive versions of a container, inspect differences, or roll back to previous versions. Containers reuse components from the preceding layers, which makes them noticeably lightweight.
- **Debugging capabilities:** Docker provides many tools that are not necessarily specific to containers, but work well with the concept of containers. They also provide extremely useful functionality. This includes the ability to checkpoint containers and container versions, as well as to differentiate between two containers. This can be immensely useful in fixing an application.
- **Multi-tenancy:** Using Docker, it is easy and inexpensive to create isolated environments for running multiple instances of app tiers for each tenant. This is possible given the spin up speed of Docker environments and its easy-to-use API, which we can use to spin containers programmatically.

Open source Docker tools for developers

The success of any software project is often measured by the ecosystem it spawns. Projects built around, next to, and on the top of the core technology enhance its power and usability, and they often move the needle forward. Docker allows IT departments to focus on applications, not virtual machines, as a standard unit of production. Looking at recent trends, there have been many projects created by the Docker community that have advanced the developer experience. Although choosing among the great contributions is really difficult, here is a roundup of the most useful open source Docker tools for developers.

Kubernetes

Kubernetes is an open source container cluster manager designed by Google for managing containerised applications in a clustered environment. The objective behind the development of Kubernetes is to provide developers a strong platform for automatic deployment, scaling and operation of application containers across clusters of hosts, providing container-centric infrastructure.

Kubernetes is regarded as portable (public, private, hybrid and multi-cloud), extensible (modular, pluggable, hookable and composable) and self-healing (with auto-replacement, auto-restart, auto-replication and auto-scaling).

It can schedule and run application containers on clusters of physical or virtual machines.

Kubernetes satisfies a number of common needs of applications for developers, like co-locating helper processes, mounting storage systems, distributing secrets, checking the application's health, replicating application instances, horizontal auto-scaling, rolling updates, resource monitoring, log access and ingestion, and support for introspection and debugging.

Dockersh

Dockersh has been developed by Yelp as part of its testing and service management infrastructure suite. Dockersh is designed to be used as a login shell on machines with multiple interactive users. When a user invokes Dockersh, it brings up a Docker container, and then spawns a new interactive shell in the container's namespace.

With Dockersh, users enter their own individual Docker containers (acting like a lightweight virtual machine), with the user's home directory mounted from the host system (so that user data is persistent between container restarts), but with its own kernel namespaces for processes and networking. This means that the users are isolated from the rest of the system, and can only see their own processes respectively, and have their own network stack which, in turn, provides better privacy between users. Dockersh eliminates the need for any of these techniques by acting like a regular shell, which can be used in `/etc/passwd` or as an SSH ForceCommand.

Dockersh tries hard to drop all privileges as soon as possible, including disabling the `sudo`, `sgid`, `raw sockets` and `mknod` capabilities of the target process.

Prometheus

Prometheus is an open source system-monitoring and alerting toolkit developed by SoundCloud. It addresses many aspects of monitoring such as generation and collection of metrics, graphing the results on dashboards, and raising alerts when anomalies occur. It works well for recording any purely numeric time series, and fits both machine-centric monitoring as well as highly dynamic service-oriented architectures.

Its components are:

- **Prometheus server:** Scraps and stores time series data
- **Client libraries:** Used for instrumenting application code
- **Push gateway:** Supports short-lived jobs
- **GUI-based dashboard builder** based on Rails/SQL
- **Special-purpose exporters**

Prometheus is designed for reliability to allow users to quickly diagnose problems. The Prometheus server is standalone, and does not depend on network storage or other remote services.

The features of Prometheus are:

- A multi-dimensional data model (time series identified by the metric name and key/value pairs)